🪔 Diwali Sales Insights Analysis

📌 Project Overview

This project analyzes Diwali sales data containing over **11,000 customer records** to uncover customer purchasing behavior, sales trends, and business insights during the festive season.

Using **Python, Pandas, NumPy, Excel, and Power BI**, the project performs data cleaning, exploratory data analysis (EDA), and dashboard creation to help businesses make data-driven decisions.

🎯 Objectives

- Clean and preprocess raw sales data.
- Analyze customer demographics and purchasing behavior.
- Identify top-performing products and categories.
- Discover regional sales trends.
- Segment customers based on spending patterns.
- Create interactive Power BI dashboards.

🛠️ Tech Stack

Programming & Analysis

- Python
- Pandas
- NumPy

Visualization

- Matplotlib
- Seaborn

Business Intelligence

- Power BI
- DAX
- Power Query

Spreadsheet Tools

- Microsoft Excel

📁 Dataset Details

The dataset contains:

- Customer ID
- Gender
- Age Group
- Marital Status
- Occupation
- State
- Product Category
- Product ID
- Orders
- Sales Amount

Total Records: 11,000+

🖌️ Data Cleaning & Preprocessing

The following steps were performed:

- Removed missing values
- Dropped unnecessary columns
- Handled duplicate records
- Corrected data types
- Standardized categorical data
- Prepared data for visualization and analysis

🔍 Exploratory Data Analysis (EDA)

Customer Analysis

- Gender-wise sales analysis
- Age group spending patterns
- Marital status insights

Geographic Analysis

- State-wise sales distribution

- Top-performing regions

Occupation Analysis

- High-spending occupations
- Purchase behavior by profession

Product Analysis

- Most sold categories
- Revenue-contributing products
- Top-performing product groups

📊 **Key Insights**

👩 **Female Customers Drive More Sales**

Female customers contributed significantly higher revenue compared to male customers.

🎯 **Target Customer Segment**

Married women aged **26–35 years** were identified as the most active buyers.

🌎 **Top Sales Regions**

Certain states generated the highest number of orders and total sales revenue.

💼 **Occupation Trends**

Customers from IT, Healthcare, and Aviation sectors demonstrated strong purchasing power.

🛍️ **Best-Selling Categories**

- Food
- Clothing & Apparel
- Electronics

📈 **Power BI Dashboard Features**

✅ Total Revenue KPIs

✅ Sales Trend Analysis

✅ Customer Segmentation

✅ Product Performance Insights

✅ State-wise Sales Analysis

✅ Interactive Filters and Slicers

📁 Project Structure

```
Diwali-Sales-Insights-Analysis/
│
├── Dataset/
│   └── Diwali_Sales_Data.csv
│
├── Notebook/
│   └── Diwali_Sales_Analysis.ipynb
│
├── PowerBI/
│   └── Diwali_Sales_Dashboard.pbix
│
├── Images/
│   ├── dashboard-overview.png
│   └── customer-analysis.png
│
├── README.md
│
└── requirements.txt
```

🚀 Installation

Clone the repository:

```
git clone https://github.com/ayushrai92840/Diwali-Sales-Insights-Analysis.git
```

Navigate to the project directory:

```
cd Diwali-Sales-Insights-Analysis
```

Install dependencies:

```
pip install -r requirements.txt
```

Launch Jupyter Notebook:

jupyter notebook

📦 Required Libraries

import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

import seaborn as sns

Install using:

pip install pandas numpy matplotlib seaborn

💡 Business Impact

This project helps businesses:

- Improve customer targeting
- Optimize inventory management
- Increase festive season sales
- Identify high-value customer segments
- Build data-driven marketing strategies

🔮 Future Enhancements

- Sales Forecasting using Machine Learning
- Customer Segmentation using Clustering
- Recommendation System
- Real-Time Dashboard Integration
- Predictive Analytics

👷 Author

Ayush Rai

- GitHub: https://github.com/ayushrai92840